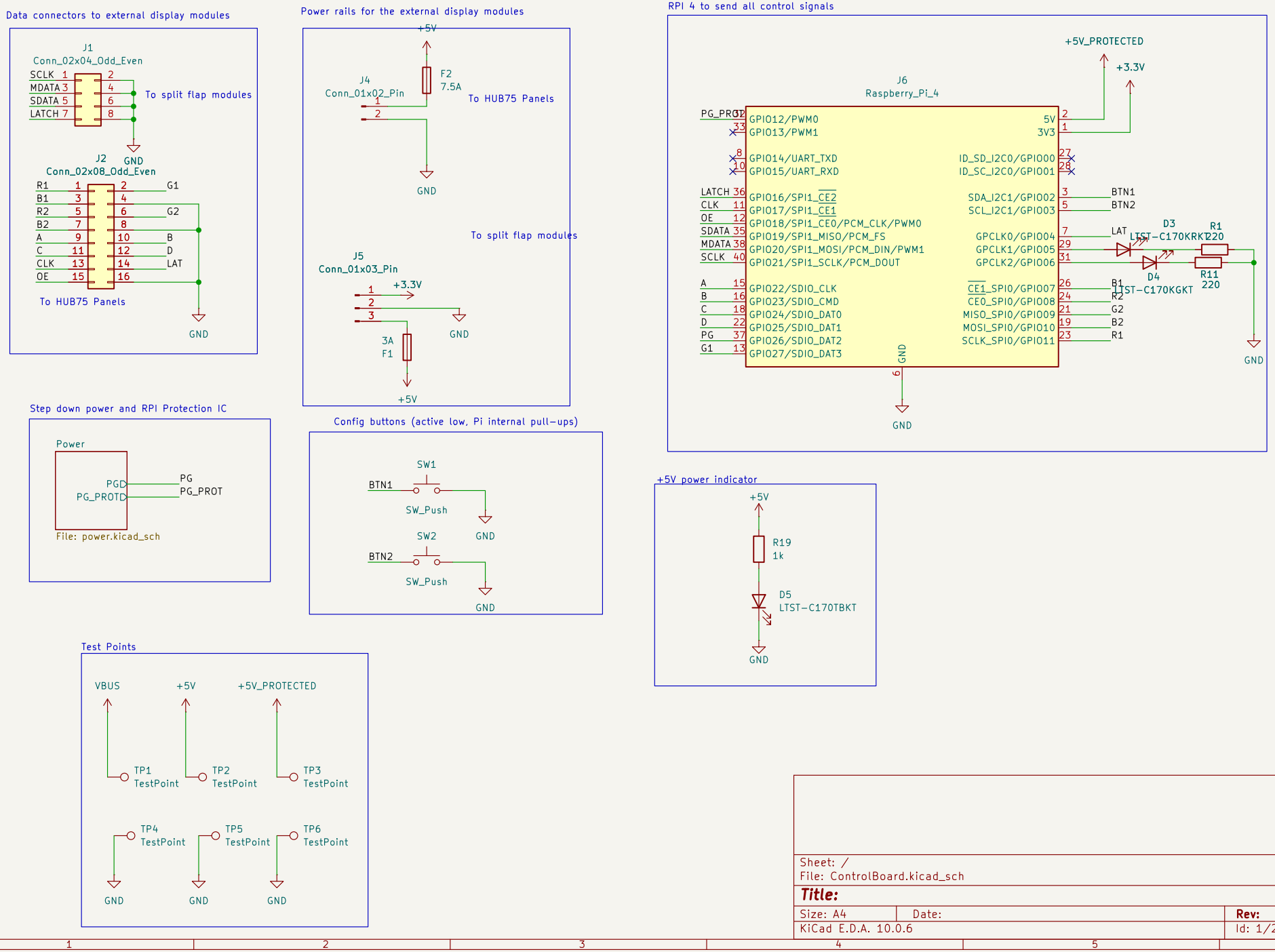
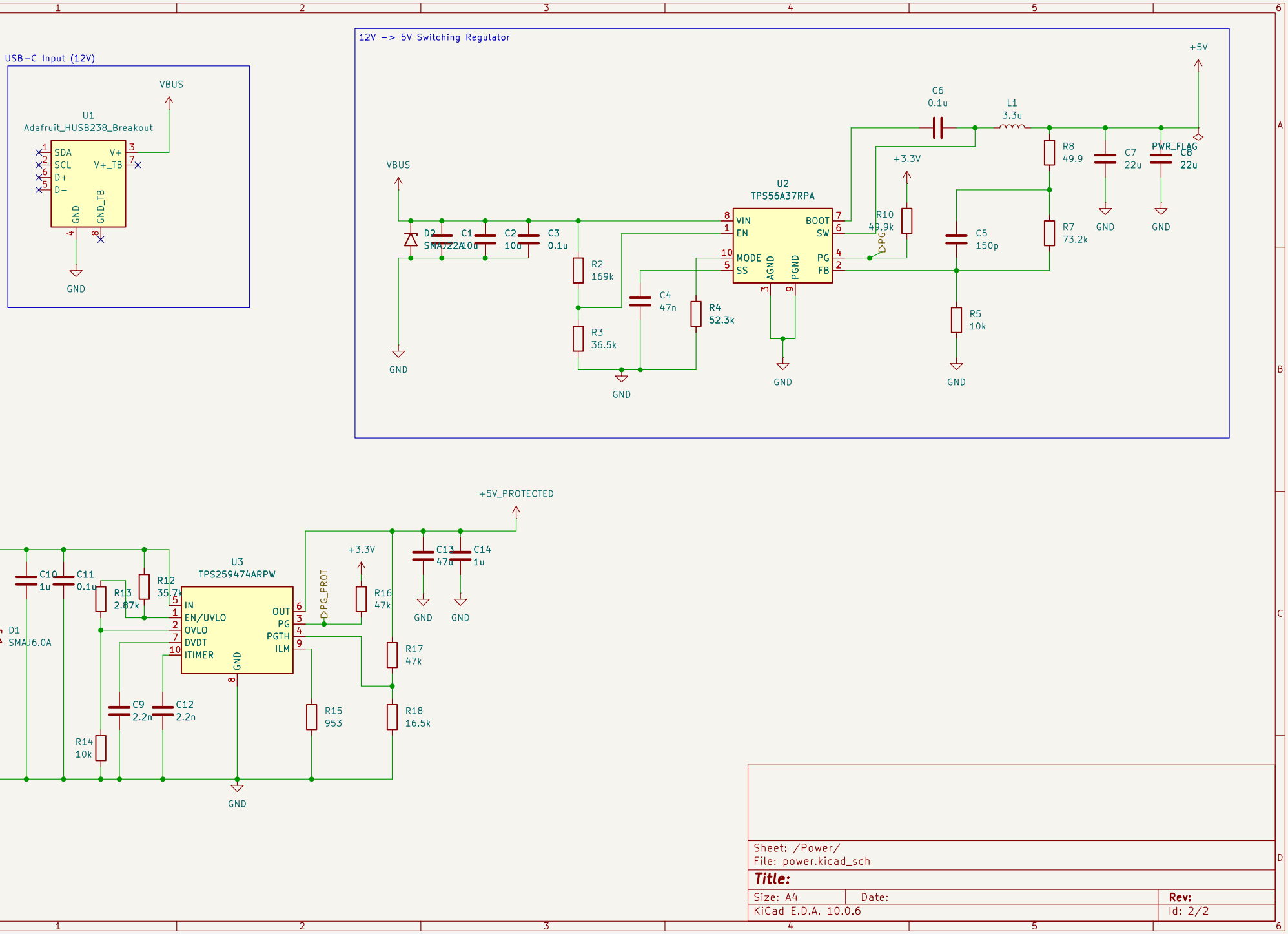

Data connectors to external display modules
J1
Conn_02x04_Odd_Even
SCLK 1
MDATA 3
SDATA 5
LATCH 7
2
4
6
8
To split flap modules
GND
J2
Conn_02x08_Odd_Even
R1 1
B1 3
R2 5
B2 7
A 9
C 11
CLK 13
OE 15
2
4
6
8
10
12
14
16
G1
G2
B
D
LAT
To HUB75 Panels
GND
Power rails for the external display modules
+5V
J4
Conn_01x02_Pin
1
2
F2
7.5A
To HUB75 Panels
GND
To split flap modules
J5
Conn_01x03_Pin
1
2
3
+3.3V
GND
3A
F1
+5V
RPI 4 to send all control signals
+5V_PROTECTED
+3.3V
J6
Raspberry_Pi_4
PG_PROT 32
33
8
10
LATCH 36
CLK 11
OE 12
SDATA 35
MDATA 38
SCLK 40
A 15
B 16
C 18
D 22
PG 37
G1 13
GPIO12/PWM0
GPIO13/PWM1
GPIO14/UART_TXD
GPIO15/UART_RXD
GPIO16/SPI1_CE2
GPIO17/SPI1_CE1
GPIO18/SPI1_CE0/PCM_CLK/PWM0
GPIO19/SPI1_MISO/PCM_FS
GPIO20/SPI1_MOSI/PCM_DIN/PWM1
GPIO21/SPI1_SCLK/PCM_DOUT
GPIO22/SDIO_CLK
GPIO23/SDIO_CMD
GPIO24/SDIO_DAT0
GPIO25/SDIO_DAT1
GPIO26/SDIO_DAT2
GPIO27/SDIO_DAT3
GND
6
5V
3V3
ID_SD_I2C0/GPIO00
ID_SC_I2C0/GPIO01
SDA_I2C1/GPIO02
SCL_I2C1/GPIO03
GPCLK0/GPIO04
GPCLK1/GPIO05
GPCLK2/GPIO06
CE1_SPI0/GPIO07
CE0_SPI0/GPIO08
MISO_SPI0/GPIO09
MOSI_SPI0/GPIO10
SCLK_SPI0/GPIO11
2
1
27
28
3
5
7
29
31
26
24
21
19
23
BTN1
BTN2
LAT
B1
R2
G2
B2
R1
D3
LTST-C170KRKT
R1
220
D4
R11
220
LTST-C170KGKT
GND
GND
Step down power and RPI Protection IC
Power
PGD
PG_PROTD
PG
PG_PROT
File: power.kicad_sch
Config buttons (active low, Pi internal pull-ups)
SW1
BTN1
SW_Push
GND
SW2
BTN2
SW_Push
GND
+5V power indicator
+5V
R19
1k
D5
LTST-C170TBKT
GND
Test Points
VBUS
+5V
+5V_PROTECTED
TP1
TestPoint
TP2
TestPoint
TP3
TestPoint
TP4
TestPoint
TP5
TestPoint
TP6
TestPoint
GND
GND
GND
Sheet: /
File: ControlBoard.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 1/2


USB-C Input (12V)
U1
Adafruit_HUSB238_Breakout
SDA
SCL
D+
D-
V+
V+_TB
GND
GND_TB
VBUS
GND
12V -> 5V Switching Regulator
VBUS
D2
SMAJ22A
C1
10u
C2
10u
C3
0.1u
GND
U2
TPS56A37RPA
VIN
EN
MODE
SS
AGND
PGND
BOOT
SW
PG
FB
C6
0.1u
L1
3.3u
+5V
+3.3V
R10
49.9k
PG
R8
49.9
R7
73.2k
C7
22u
C8
22u
PWR_FLAG
C5
150p
R2
169k
R3
36.5k
C4
47n
R4
52.3k
R5
10k
GND
+5V
+5V_PROTECTED
D1
SMAJ6.0A
C10
1u
C11
0.1u
R13
2.87k
R12
35.7k
U3
TPS259474ARPW
IN
EN/UVLO
OVLO
DVDT
ITIMER
GND
OUT
PG
PGTH
ILM
PG_PROT
+3.3V
R16
47k
C13
47u
C14
1u
R17
47k
R18
16.5k
R15
953
C9
2.2n
C12
2.2n
R14
10k
GND
Sheet: /Power/
File: power.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 2/2